

SHAHZEB UMER

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 1+ years of professional experience building and maintaining production web applications in Node.js/Express and React/Next.js. Experienced in designing REST APIs, integrating third-party and AI/LLM APIs, working with relational databases, and deploying to cloud infrastructure. Comfortable owning backend services end-to-end in fast-paced, startup-style environments.

EDUCATION

Bachelor of Computer Science, NUCES-FAST University 2021 - 2026

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Software Engineering, Web Development, Operating Systems, Artificial Intelligence, Object-Oriented Programming

Pre Engineering, KIPS College

2019 - 2021

TECHNICAL SKILLS

Programming Languages	JavaScript, TypeScript, Python, C++, C#, SQL
Backend Technologies	Node.js, Express.js, REST APIs, FastAPI, .NET Framework
Frontend Technologies	React.js, Next.js, React Native, HTML5, CSS3, Tailwind CSS
Databases	PostgreSQL, MySQL, MongoDB, MS SQL Server
AI & API Integration	OpenAI-compatible & Google Gemini APIs, RAG, WebSockets, retry & logging
DevOps & Cloud	Docker, Git, CI/CD, AWS, object storage, Nginx, Kubernetes
Tools & Practices	Strapi CMS, LaTeX, Agile/Scrum, Performance Optimization

PROFESSIONAL EXPERIENCE

Full Stack Developer Dec 2024 - Present

Directorate of ICT, Allama Iqbal Open University (AIOU) Islamabad, Pakistan

Leading distance learning institution serving 400,000+ students nationwide

Islamic Research Index (IRI) - iri.aiou.edu.pk

- Built and deployed a Next.js 14 research-indexing platform hosting thousands of academic papers
- Optimized database queries and added caching to improve page-load performance
- Automated content-management workflows, reducing manual indexing effort

HR Management System - AIOU (*internal*)

- Built a full-stack HR Management System (Node.js/Express + Next.js) with REST APIs and PostgreSQL models for attendance, payroll, evaluations, and employee lifecycle
- Implemented role-based access control and stored employee documents and media in cloud object storage
- Automated manual HR workflows, improving administrative efficiency and data accuracy

AIOU Bookstore E-commerce Platform - bookstore.aiou.edu.pk

- Built and maintained an e-commerce platform with integrated inventory management for university students
- Implemented secure payment processing and a modular architecture (online sales, POS, warehouse, email notifications)
- Synchronized inventory between the online store and physical outlets to reduce stock discrepancies

PROJECTS

BumbleTap — Browser Automation Extension [bumbletap.com] [Chrome Web Store]

Technologies: Chrome Manifest V3, JavaScript, Next.js, Cloudflare Workers & R2, Vitest

Architected and published a Chrome extension that binds arbitrary keystrokes to DOM actions on any website, composed of two primitives: single-key bindings (element invocation, text entry, sandboxed user JavaScript) and multi-step Auto-Actions supporting conditional branching, waits, and variable extraction. Engineered a resilient element resolver that identifies targets via multi-representation consensus and shadow-DOM traversal, sustaining automations across site redeloys that invalidate conventional CSS selectors. Orchestrated privileged execution across Chrome's isolated, main, and user-script worlds through a cross-world messaging bridge, enabling CORS-exempt requests without compromising a strictly client-side design — no backend, no telemetry. Authored the companion statically-exported Next.js site and engineering blog, deployed on Cloudflare's edge.

Lumen — AI Video Understanding Platform [GitHub]

Technologies: Next.js, React, FastAPI, WebSockets, OpenAI-compatible & Gemini APIs, ChromaDB

Built a full-stack platform that transcribes, summarizes, and answers questions about videos using LLM APIs for text, audio transcription, and vision. Implemented a real-time streaming pipeline over WebSockets with live progress events, and added retry and structured logging around every external API call. Developed a Retrieval-Augmented Generation (RAG) Q&A feature using a vector store with embedding-based retrieval and reranking to produce grounded, timestamp-cited answers, plus Markdown and media export.

Customer Support Automation Platform [GitHub]

Technologies: Next.js, Python, SQLite, Machine Learning

Built a multi-tenant support platform with ML-based ticket routing and NLP sentiment analysis to automatically categorize and prioritize incoming tickets. Implemented automated response suggestions and a dashboard for tracking ticket status and team performance.

LEADERSHIP & ACTIVITIES

- **Technical Content Creator** - Author an ongoing engineering blog on browser internals, automation, and web architecture, alongside technical writing and professional insights on LinkedIn.